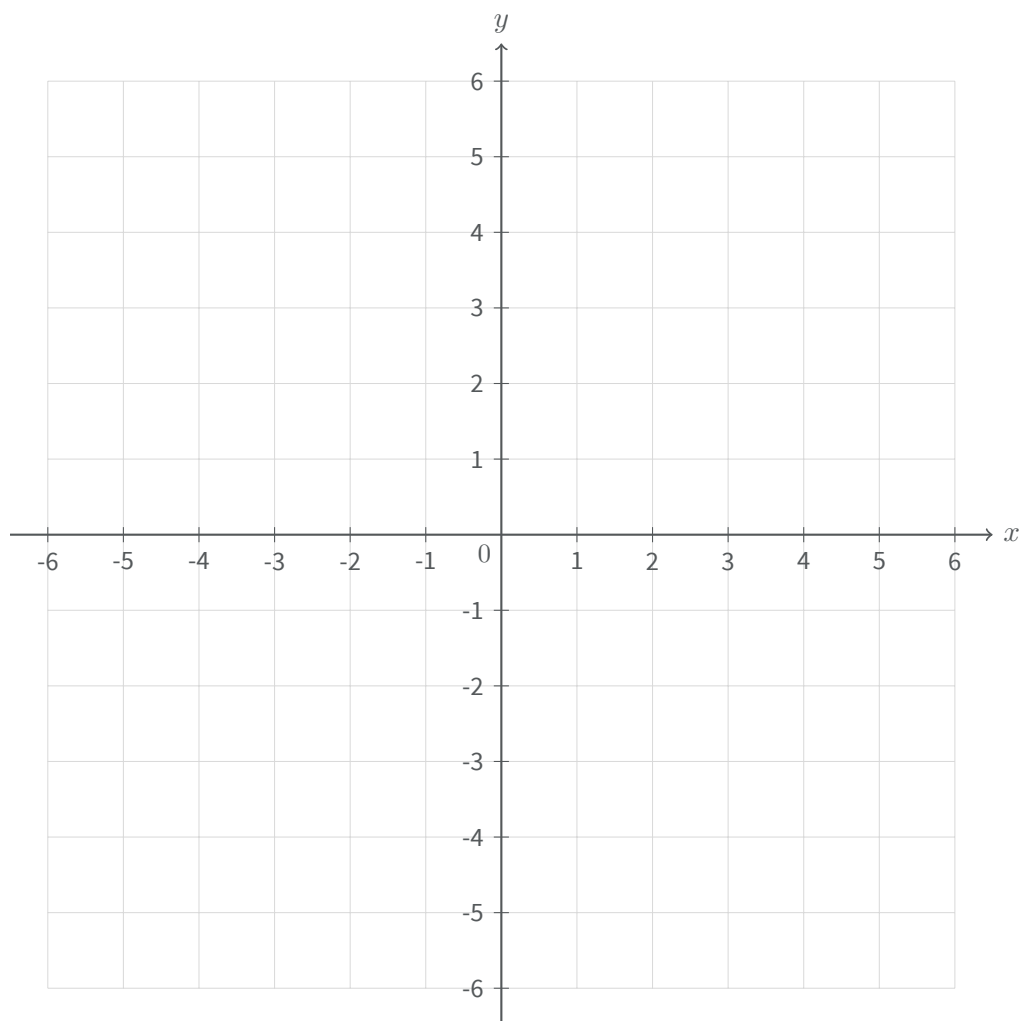


M^5 Day 10: Rational functions

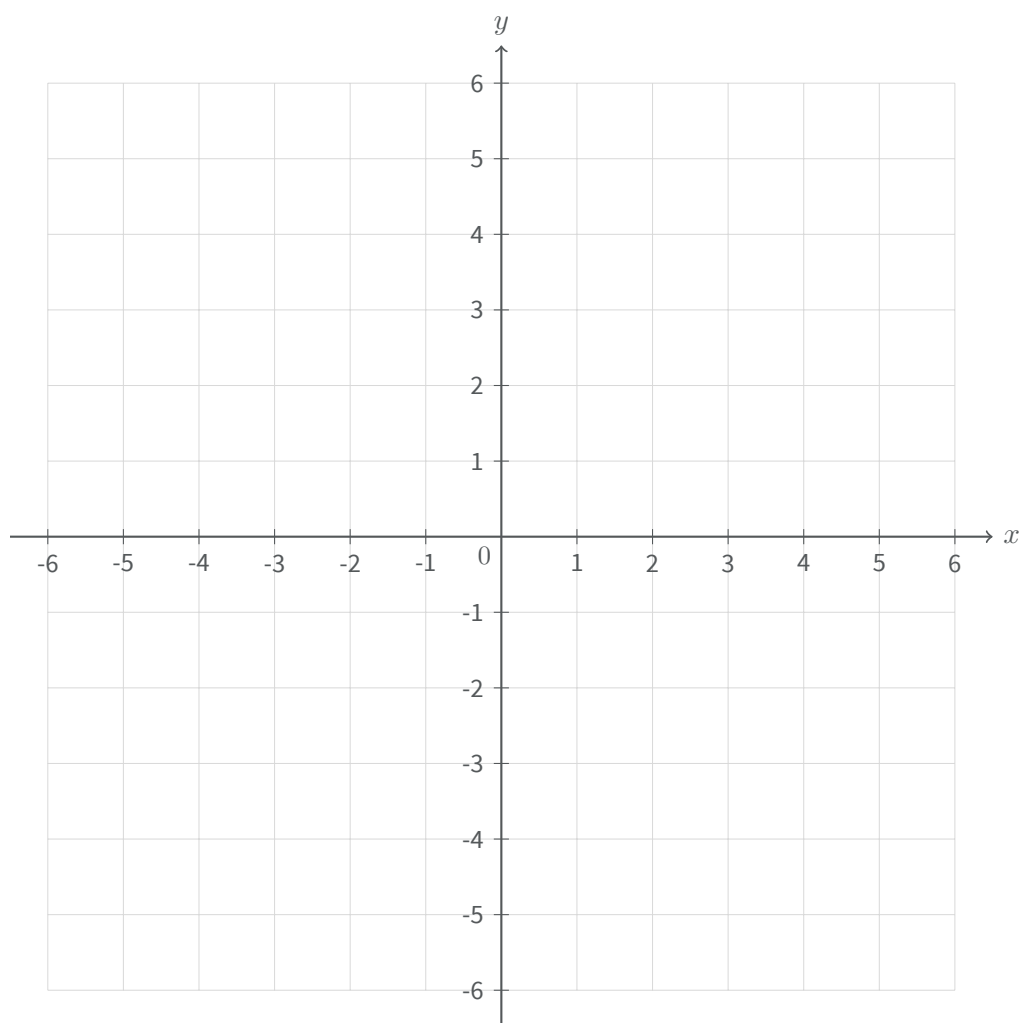
Name: _____

1. Sketch the following rational functions:

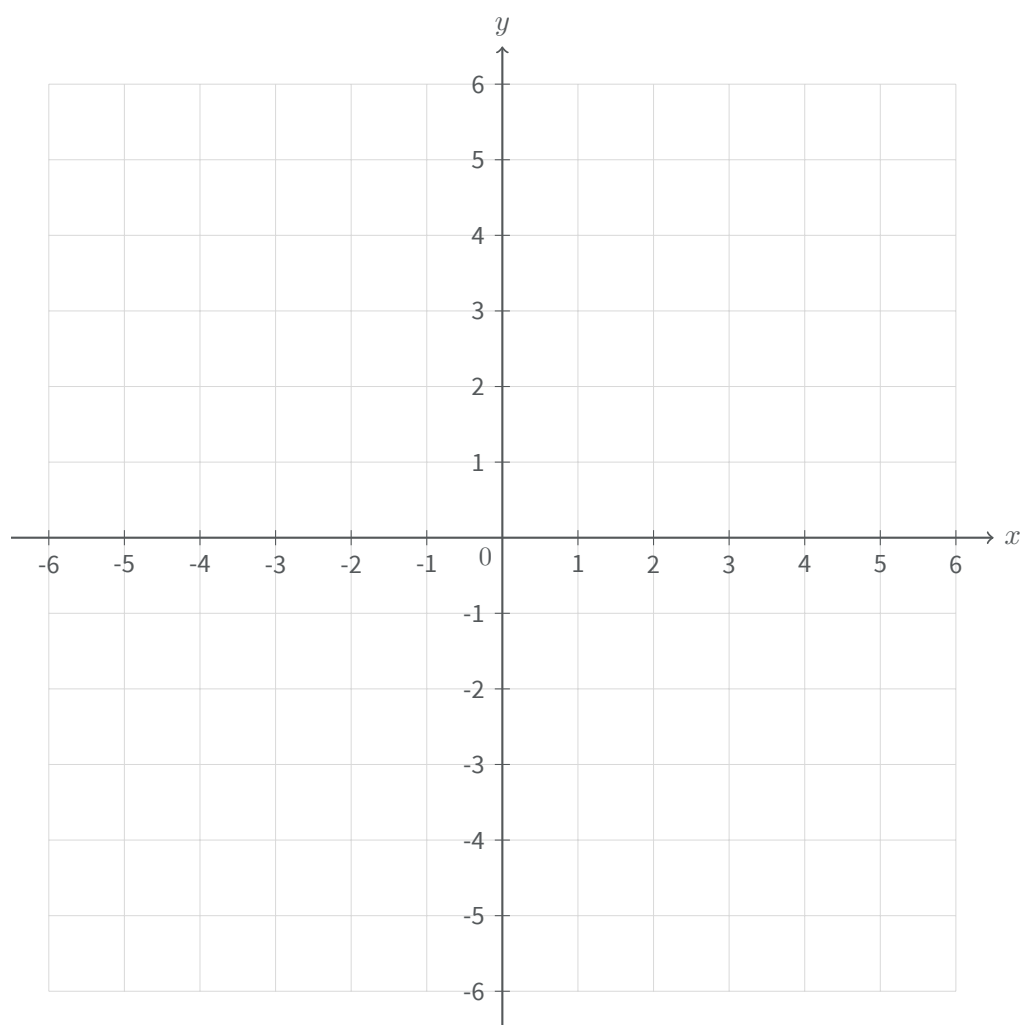
a. $y = \frac{(x - 2)(x + 2)}{(x - 1)}$



b. $y = \frac{1}{(x+1)(x-2)}$



c. $y = \frac{(x-2)(x+2)}{(x-1)}$



2. Plot the values of x that satisfy $\frac{(x-3)^2(x+1)}{(x+2)^3} \geq 0$ on the real number line.

3. Plot the values of x that satisfy $\frac{1}{x^2-1} < 0$ on the real number line.

4. Plot the values of x that satisfy $\frac{x^2-2x-2}{x-1} < 0$ on the real number line.